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Promoting validity of design principles for teaching through educational design research

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ABSTRACT

Educational design research (EDR) needs to further clarify how it leads to valid theoretical results. In this article, we aim to contribute to the development of EDR by suggesting an explication of what validity of design principles for teaching entails, and how the characteristics and phases of EDR can be employed to increase such validity. Our suggestions are based on our own adaptation of EDR within a research program focusing on teaching that promotes students' mathematical reasoning, informed by discussions and elaborations of EDR and validity of causal inferences. Our results elaborates existing causal validity types in relation to design principles for teaching and adds two new validity types: relative benefit and guiding power. We also describe the importance of employing multi-coder level-raising analysis using an intermediate framework in collaboration with teachers to develop theory over iterations to increase not only explanatory power but also predictive and guiding power.

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One of the primary goals of educational research is to develop knowledge that guides instructional decisions (Alexander, 2013). Educational design research (EDR) has been argued to be better suited for studying educational settings and to produce more useful results than, for example, single randomized controlled experiments (Brown, 1992; McKenney & Reeves, 2018). It has, however, also been criticized for lacking a basis for warrant for their claims (Cobb et al., 2015; De de, 2004; Kelly, 2004; Shavelson et al., 2003). Addressing this critique is complicated by the fact that EDR comprises a variety of approaches and aims for multiple goals. It has evolved from multiple methodological roots in curriculum development, didactical engineering and teaching experiments (Prediger et al., 2015). In the field of mathematics education research, design of tasks has played a central role since its formation in the late 1960s, first inspired by psychological theories and increasingly linked to classroom practice, with theoretical and methodological developments in, for example, France, the Netherlands, and Britain (Kieran et al., 2021). Origins of teaching experiments in mathematics date further back to the 1920s in the Soviet Union (Kieran et al., 2021) and the 1930s in the United States (Thiele, 1938). It aims to produce both practical and theoretical outcomes, as well as different types of theoretical outcomes: descriptive, explanatory, predictive, and prescriptive (Brown, 1992; McKenney & Reeves, 2018). Descriptive and explanatory theories are used after the fact to give accounts and explanations of what has occurred, while predictive theory is used before the fact to predict what is going to happen. Prescriptive theory gives recommendations for what to do and does therefore rely on the three other types of theory.

Substantial contributions have been made to the development of EDR as a methodology for explanatory theory development, especially in mathematics education by Cobb and colleagues

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(Cobb & Gravemeijer, 2008; Cobb et al., 2003, 2015, 2017; Gravemeijer & Cobb, 2006). In this article, we aim to contribute to the development of EDR as a methodology by focusing on a particular type of prescriptive theoretical outcome from EDR: design principles for teaching. We aim to explicate what validity of design principles for teaching entails, and how EDR can be employed to increase validity of design principles for teaching.

Design principles as prescriptive theoretical outcomes of EDR

One way that EDR takes on the challenge of producing more useful results for practice is that it aims to develop theory for prescriptive purposes, that is, a theory that answers questions on the form "If one wants Y, what should one do?" Specified for teaching, these questions could take the form: "If one wants one's students to learn Y, what should one do?" Such questions are commonly asked by teachers, but they are not common as research questions. One reason for this might be that researchers do not see such questions as researchable or argue that science is about what is and not about what one should do (Wecker, 2013). Nonetheless, such questions are often discussed as implications of a study (Robinson et al., 2013), indicating that many researchers believe that their results have bearing on such questions, even if the results do not answer them per se.

Answers to questions on the above form are prescriptive claims (Wecker, 2013): If one wants one's students to learn *Y*, one should do *X*. However, prescriptive claims alone may not provide sufficient answers for teachers, for at least two reasons. First, the function of teaching is often not located in the mere performance of actions but in doing the action with a specific purpose. For example, Melhuish et al. (2024) found that the purpose of teachers' why-questions, rather than the questions' form or frequency, explained variation in students' classroom activity. Second, a prescriptive claim can scarcely capture the diversity of situations teachers will encounter in the classroom, to which the claim needs to be adapted. The teacher's task can never be reduced to simply following a given teaching material but always entails interpretation, judgment, and adaptation (Prediger et al., 2015). For these reasons, prescriptive claims provide better guidance if they are accompanied by arguments explaining how doing *X* leads to students learning *Y* so that the teacher can use these explanations to evaluate and adapt *X* to various situations. van den Akker (2010) has described the combination of prescriptive claims and arguments as design principles and suggested the following format for design principles for the design of interventions:

> If you want to design intervention *X* [for purpose/function *Y* in context *Z*], then you are best advised to give that intervention the characteristics $C_1, C_2, \ldots, C_m$ [substantive emphasis] and to do that via procedures $P_1, P_2, \ldots, P_n$ [methodological emphasis] because of theoretical arguments $T_1, T_2, \ldots, T_p$ and empirical arguments $E_1, E_2, \ldots, E_q$.

A design principle for the design of elements of teaching – such as tasks, support for students or orchestration of whole-class discussions – would need to be formulated slightly differently, but one may note a direct correspondence between van den Akker's purpose or function *Y* and the goal for students' learning *Y*, and a correspondence between teachers doing *X* and the intervention *X* with characteristics $C_1, \ldots C_m$ given via procedures $P_1, \ldots P_n$. One may also note that van den Akker (2010) includes a specification of context, indicating that one should specify the limits for the recommendations made. This could possibly also be done in other ways, for example, by further specification of the learning goal. In any case, to be meaningful as a recommendation for others, a design principle cannot be restricted to the exact context that was empirically studied, implying that the theoretical and empirical arguments need to include a basis for determining how the design principle can be generalized from the contexts studied to new contexts.

Characteristics and phases of EDR as means for theory development

EDR has been described in terms of characteristics and phases in various ways. Descriptions of the EDR process range from focusing on phases involved in the devising, testing, and evaluation of designs (e.g., McKenney & Reeves, 2018), to descriptions of the whole research process, from reviewing existing literature and explicating research questions (e.g., Bannan-Ritland, 2003; Easterday et al., 2018). Three characteristics common to most descriptions are that EDR is *collaborative*, *interventionist*, and *iterative* (Cobb et al., 2003, 2015, 2017; McKenney & Reeves, 2018).

Collaborative means that the EDR process is, to varying degrees, conducted in collaboration between researchers and practitioners. While the necessity and value of collaboration in EDR is often stated, the arguments for these claims are seldom explicit. When they are, they mostly focus on practitioners' contribution in formulating the problem and designing the practical solution to it (see, e.g., McKenney & Reeves, 2018), leaving the questions of whether and, if so, how collaboration aids the development of theoretical outcomes of EDR unanswered.

Interventionist means that a design is introduced to a context and the effects studied. This highlights that EDR is an experimental, rather than naturalistic, approach. This characteristic is reflected in three phases that are commonly included in descriptions of the EDR process: hypothesizing empirical experimentation and analysis. The hypothesizing phase entails preparation for an experiment by formulating hypotheses about how properties of elements of teaching will affect students' learning. The empirical experimentation phase entails empirical testing of a design, meant to embody the theoretical hypotheses. The analysis phase entails evaluation of and reflection on the empirical tests.

Iterative means that results from previous experiments are used for revising the design before use in subsequent experiments, which allows for development of designs, and possibly theory. However, many descriptions of iterating within EDR do not include theory development at all. Easterday et al. (2018, p. 131) state that "theory and interventions drive each other in complex, iterative ways" but do only describe how the phases develop designs. Other models place iterations within the testing phase and analysis after all iterations are conducted, restricting the iterative component to the development of designs (e.g., Bannan-Ritland, 2003; Hoadley & Campos, 2022). Cobb and colleagues have an explicit focus on EDR for the purpose of developing the theory. They stress that the need for theoretical orientation in iteration over all three types of phases to "put theory in harm's way," that is, to use empirical experiments to test, evaluate, and revise conjectures about how teaching can support learning (Cobb et al., 2003). However, they focus development of explanatory theory in the form of local instructional theories, explaining how a set of domain-specific activities promoted specific learning processes and outcomes in the experiment for example, the idea of statistical distribution or addition and subtraction up to 100 (Cobb & Gravemeijer, 2008; Gravemeijer, 2004). The focus on explanatory theory may be the reason why theory development is placed mainly in a phase after all iterations, called retrospective analysis (Cobb et al., 2003, 2015, 2017; Gravemeijer & Cobb, 2006).

In sum, the descriptions of EDR often focus more on how the characteristics and phases of EDR contribute to the development of practical solutions than how they contribute to the development of theory. It has been stated that the role of theory in EDR is complex and dynamic (Prediger et al., 2015) and that it is crucial to "put theory in harm's way" (Cobb et al., 2003), but how, why, and under what circumstances the key characteristics and phases of EDR develop prescriptive theory and, in particular, design principles for teaching have not been described explicitly.

Aim

Kelly (2004) stresses that for EDR to be a methodology, there must be arguments for how the methods lead to demarcation between sound and unsound claims. However, EDR has multiple purposes and can thus produce claims of different kinds. We aim to contribute to the explication of EDR as

a methodology by focusing on one type of prescriptive theoretical outcome that EDR can produce: design principles for teaching. More specifically, we aim to explicate:

(1) what validity of design principles for teaching entails, and
(2) how key characteristics and phases of EDR can be employed to increase the validity of design principles for teaching.

Context and approach to address the aim

Our contribution stems from our own use and adaptation of EDR methods within the research program Learning by Imitative and Creative Reasoning (LICR). LICR spans 25 years and over 50 studies and aims to clarify the consequences, characteristics, and prerequisites of students' imitative and creative reasoning in mathematics. The program began with observational studies of characteristics of student learning difficulties in mathematics, focusing on the difference between imitation of procedures and own creation of solutions. One of the results was a research framework for imitative and creative mathematical reasoning characterizing why and how learning by imitation leads to superficial knowledge and explaining how deeper learning can be obtained by constructing one's own task solutions (Lithner, 2008). This framework guided subsequent laboratory studies of task designs enhancing imitative or creative mathematical reasoning. By applying behavioral methods, eye-tracking, and fMRI brain imaging, these studies have showed how creative task design can lead to more effective learning (Lithner, 2017).

In the early 2010s, the program turned to studies on how to support students' own creative reasoning in regular classrooms. In line with international research, LICR studies had shown that Swedish mathematics teaching (Boesen et al., 2014) and mathematics textbooks world-wide (Jäder et al., 2020) were dominated by imitative teaching designs focusing students' work on routine tasks, leaving little room for creative mathematical reasoning. A main question for the program was

> If one wants one's students to develop mathematical competence, what kind of tasks should one give to students, and what kind of support should one provide if a student needs help during task solving?

The reason for the focus on tasks and teacher support was that they are key elements of mathematics teaching which we conjectured could have substantial impact on students' reasoning. Furthermore, while being perceived as a challenge for teachers, there was less existing research on teacher support during task solving than, for example, follow-up discussions on solution strategies.

At this time, the design principle format suggested by van den Akker (2010) was found to be the clearest expression of answers to questions of the type we aimed to investigate, and it was therefore used as a starting point. However, van den Akker (2010) does not specify what the arguments of design principles can or should be, and as our aim was to develop design principles for teaching, we therefore combined his ideas with the idea of hypothetical learning trajectories (Simon, 1995). This allowed us to specify the arguments of design principles for teaching as relating properties of three components: elements of teaching (*E*), learning processes (*L*), and attainment of a learning goal (*G*). This idea was later informed by other similar models, such as Sandoval's (2014) conjecture mapping and Maxwell's (2004) causal explanation, resulting in the format for design principles for teaching presented in Figure 1. In the rest of this article, we focus design principles on this format.

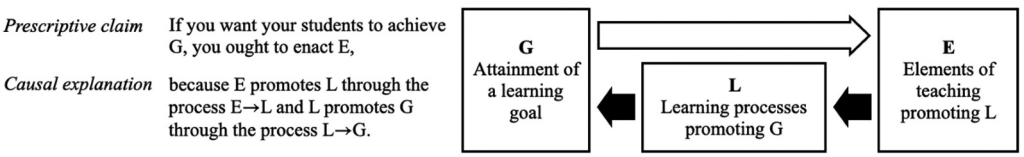


Figure 1. A proposed format for design principles for teaching.

The model is to be seen as minimal in the sense that it specifies the theoretically necessary components of a design principle for teaching. Depending on one's aim and methodological choices, one may also include additional components, such as specifications of what groups of students, situations, or other contextual factors the design principle is limited to.

Two examples of initial, tentative design principles (TDPs) from the LICR program, consisting of a prescriptive claim and a causal explanation supporting the claim, are:

TDP 1: a tentative design principle for tasks

- **Prescriptive claim**: If the goal is for students to develop mathematical competence, the teacher should give the students problems, that is, tasks where the students do not have access to a solution method beforehand.
- **Causal explanation**: Because mathematical problems promote creative mathematical reasoning (CMR) through their absence of an available (given or recalled) solution method, that is, there is no other way for the student to solve problems than by means of their own reasoning (Lithner, 2008), and CMR leads to higher levels of mathematical competence (Schoenfeld, 1985; Jonsson, 2014) than applying available solution methods.

TDP 2: a tentative design principle for teacher support during problem solving

- **Prescriptive claim**: If the goal is for students to develop mathematical competence and a student needs help while working on a problem, the teacher should use formative assessment to select feedback based on the student's specific difficulty,
- **Causal explanation**: Because formative assessment is defined as gathering, interpreting, and using information about the student's achievement to make decisions that are likely to be better for the student's learning than decisions made in the absence of that information (Black & William, 2009). Feedback that is adapted to the student's specific difficulty is likely to better support and leave more responsibility for the student's CMR (Brousseau, 1997), and CMR leads to higher levels of mathematical competence (Schoenfeld, 1985; Jonsson, 2014).

From these examples, one may note that the causal explanations included in design principles for teaching are more elaborate than a mere causal description stating that E leads to G, which is what one usually tests in single experiments. At the same time, it is evident that TDP 1 and 2 are not sufficient answers to the question above, since many of the key terms are not defined. For example, TDP 2 does not give any guidance for what information the teacher should gather, how that information should be used to diagnose the student's difficulty, and what feedback works for what difficulties. One may also note that the absence of specifications regarding contexts in TDP 1 and 2 implies that they hold for all students and situations. One may question whether the referenced studies support such generality. It is thus clear that TDP 1 and 2 require development.

Figure 1 was used as a basis for formulation of a process model for development of design principles. This model was tested and revised as guidance for and a description of our own EDR process and as a tool for building arguments for our results, through continuous iteration, reflection, and revision. The reflection was guided by the questions "Does the model accurately describe what we do?," and "Is what we are doing contributing to the development of valid design principles?" The reflection was also informed by the research literature. When we started the work, we found no existing explicit description of validity of design principles of the format we were working with, and we therefore searched for descriptions of validity of causal and prescriptive claims, since a design principle on our format includes a prescriptive claim and at least two causal claims about how *E* leads to *L* and *L* leads to *G* (Figure 1).

For validity of causal claims, we started from Shadish et al.'s (2002) four types of validity for generalized causal inference: *statistical conclusion validity*, *internal validity, construct validity*, and

external validity. Predecessors of this typology (Campbell & Stanley, 1963; Cook & Campbell, 1979) focused validity of causal claims based on statistical methods for analyzing experiments grounded in a probabilistic approach to causality. It has since been informed by critique and contributions from researchers with other perspectives on causality and validity, including qualitative researchers such as Eisenhart and Howe (1992) and Maxwell (2002). The typology suited our purposes in both its specificity and generality. It is specific in the sense that it concerns generalized causal inference, that is, it is not for research in general or for explaining causal mechanisms in singular cases. This made it a suitable starting point for our purposes as prescriptive claims require a basis in a general causal claim. Claiming that "If you want your students to achieve *G*, enact *E*" can never be valid if *E* does not contribute to students achieving *G* with some generality, which allows for generalization from the contexts studied to new contexts (Martin, 2011). The typology is general in the sense that it does not rely on a single definition of causality but rather views different conceptualizations as contributing to the understanding and study of causal relationships. It adopts a view that multiple methods and research designs are needed to develop causal descriptions and explanations. This made it suitable as a basis for thinking about iterations and experiments, which are key characteristics of EDR.

For validity of prescriptive claims, we found less direct guidance. Discussions on recommendations for practice within the research community contributed to the idea that while validity of causal inferences is necessary for validity of prescriptive claims, it cannot be sufficient (Wecker, 2013). Even if enacting *E* leads to more students achieving *G* than alternatives to *E*, it is not always warranted to state that "If a teacher wants their students to achieve *G*, they should enact *E*." To do so, one must consider other effects of *E*, such as the effort required by the teacher and additional consequences for the students, in comparison to the effects on the attainment of *G* (Wecker, 2013). This resulted in the fifth validity type *relative benefit*.

An idea from DiSessa and Cobb (2004) was used to explicate an additional reason why not every causal explanation can serve as the basis for a prescriptive claim. Causal models are sometimes discussed in terms of their explanatory and predictive power: how well do they explain what happened after it happened, and how well do they predict what will happen in the future. While these powers are important for prescriptive uses of theory, they are insufficient, since it should not only predict outcomes of future actions but provide guidance for them. As stated by DiSessa and Cobb (2004): "theory must do real design work in generating, selecting and validating design alternatives at the level at which they are consequential for learning." In other words, design principles should not only be valid in the sense of being well founded, but also in the sense of being meaningful and producing the desired results. These ideas resulted in a sixth validity type: *guiding power*.

Validity of design principles

In this section, we will elaborate six validity types for design principles for teaching. The first four are based on the validity types for generalized causal inference described by Shadish et al. (2002) and primarily concern the causal explanations within design principles. As these four validity types are often associated with testing causal descriptions through randomized experiments and statistical methods, we put emphasis on their meaning in relation to developing causal explanations through EDR. The first two concern the validity of the causal explanation in relation to the studied cases, while the next two concern the validity of the causal explanation more generally. The last two types are argued to be required for it to be valid to formulate prescriptive claims based on valid causal explanations. We will use TDP 1–2 to exemplify the meaning of the six validity types.

Covariation validity, or statistical conclusion validity

Statistical conclusion validity is the validity of inferences about whether there is a covariation between cause and effect in the studied cases, and if so, how strong that covariation is (Shadish et al., 2002).

This type of validity is usually discussed in relation to testing causal descriptions using statistical measures of significance. Statistical methods depend on conditions that are often not fulfilled in educational research, such as independence of variables, firm control of conditions, and random allocation. They are also restricted to inferences about group-level effects.

The issue of covariation is nonetheless relevant for causal explanations in design principles. For example, when developing TDP 2, it is crucial to know if properties of students' reasoning covary with properties of teachers' support, and if levels of mathematical competence covary with the kind of reasoning students use. If variations observed are due to chance or if an existing covariation is very weak, there might not be any basis for valid prescriptive claims about how to support students' CMR. The existence of covariation and the risk that observed variation is due to chance can be discussed in relation to research design and findings, even if statistical tests are not used (Shadish et al., 2002). For example, that one type of feedback leads to students engaging in CMR in 70 of 100 cases where it was used, while another type of feedback only did so in 5 of 100 cases, can be discussed as support for covariation between type of feedback and students' CMR. To stress this, we use the term covariation validity rather than the more common statistical conclusion validity below.

Internal validity

Internal validity concerns the causal claims made regarding the studied cases. For design principles, this includes the two causal claims in the causal explanation: did E cause L by the process $E \rightarrow L$ and did L cause G by the process $L \rightarrow G$ in the studied cases? It is thus related to the explanatory power of the causal model underlying a design principle. For TDP 2, a crucial question is whether the student's re-engagement in CMR was caused by the teacher's support, and not by something else. In general, "cause" must be understood as contributing to the outcome in a meaningful way, as no teaching, no matter its properties, can guarantee learning to happen. The main point is that learning is not a result only of what the teacher does, but also of factors outside of the teacher's control, such as the student's will and affective state and what goes on inside and outside the classroom. We do, however, also know that in complete absence of teaching in any form – books, videos, or someone setting up situations or materials – students will learn very little mathematics. This indicates that teaching is a key causal factor for learning. One possible way to handle the indeterministic relationship between teaching and learning is to think of this relationship as probabilistic (Morrison, 2009), that is, seeing the teacher's actions as increasing the likelihood of learning to happen or not. Another way is to analyze each case – both when learning occurred and when it did not – to build explanations of what happened and what role the teacher's actions played. Arguably, one is best advised to do both, since these approaches can support each other. That one feedback type leads to CMR more often than another is one kind of argument for recommending the first type. That the cases where the first type of feedback led to CMR include evidence of the student using the feedback in their reasoning, while the cases where the second type led to CMR did not; and that the cases where the first type of feedback did not lead to CMR are cases where the student did not engage in any reasoning no matter what the teacher said, while the cases where the second type did not lead to CMR showed the student trying to use the feedback but ending up doing something else than CMR is another kind of argument. Only relying on the first kind of argument is not that helpful for developing better suggestions for feedback than the types tested or better descriptions of the salient features of more functional feedback. Since there are innumerous possible feedback formulations, it is neither feasible to repeatedly compare pairs of feedback types. However, only relying on the second kind of argument, disregarding any quantitative descriptions of one's data, risks producing recommendations that only work in very rare cases.

To ensure internal validity, one has to exclude other possible causes, for example, that the presence of an observing researcher made the students try harder and thus more engaged in CMR, and the possibility that the causal effect is converse, for example, that the teachers chose to use a specific type of

support only with students they knew always engage in CMR and another type of support with students who never do so. It is clear from these examples that while making sure that the *E* precedes *L*, which in turn precedes the measurement of *G*, is important (Shadish et al., 2002), it is not enough to establish internal validity for design principles for teaching.

Construct validity

Construct validity concerns the validity of inferences about the particulars of the empirical data and the theoretical constructs that are used to describe them (Shadish et al., 2002). Constructs can be seen as sets of properties that are given names. We view the three components *G*, *L*, and *E* as constructs, but also the students, the teachers, and the context for which the design principle claims to hold. This means that, for example, *E* should be thought of as a type of element defined by certain properties, not a particular element existing in the real world. A key construct in both TDP 1 and 2 is CMR: reasoning that has the properties novelty, plausibility, and mathematical foundation (Lithner, 2008). A key construct in TDP 2 is formative assessment: a teaching practice that entails eliciting, interpreting, and using evidence about students' achievement to make better instructional decisions (Black & Wiliam, 2009).

Shadish et al. (2002) describe two main ways in which construct validity can be violated: (a) the particulars only have a subset of the construct properties, meaning that the construct is more specific than the data, and (b) the particulars all have additional properties, meaning that the construct is more general than the data. In both cases, there is a mismatch between the data and the specificity of theory. In TDP 1 and 2 above, several constructs must be examples of (b). For example, "formative assessment" does not specify what information to gather, what specific difficulties exist and how one can identify them using the gathered information, and what kind of feedback provides support while letting the student keep the responsibility for constructing the solution. In TDP 1, "problems" would be an example of (b) if the empirical data were restricted to tasks on linear geometrical patterns, as would "students" in both TDP 1 and 2, if the data were restricted to a particular age. For TDP 2, it would also be important to specify in which contexts it holds, for example, considering if it is valid to recommend the teacher to base their support only on the student difficulty when there is little time left on the lesson, or the student is very frustrated.

When the aim is to develop causal explanations for teaching and learning, rather than only to test causal descriptions, construct validity can be violated in an additional way: (c) the properties defining the construct are not the properties salient for the causal inference. Even if *E* did cause a change in the learning of the students and *E* is described at an appropriate level in relation to the particulars, it is often possible to describe *E* in alternative ways. In themselves, these alternative descriptions may all be true in the sense that they reflect common properties of the elements studied. However, all these descriptions may not capture the properties that affected students' learning equally well. If the constructs do not capture the salient properties, the design principle will not highlight what is important to achieve *G* and cannot be valid.[1]

When testing causal descriptions, it is paramount to clearly explicate the constructs and carefully select instances that match those constructs before the data are collected (Shadish et al., 2002). When the aim is to develop design principles, one should expect revision, and even innovation, of new constructs during the process (DiSessa & Cobb, 2004). Construct validity should thus be seen as a result of, rather than a prerequisite for, experiments in EDR.

External validity

External validity concerns the extent to which causal effects hold over the variation within the study and for generalizations to cases beyond the study (Shadish et al., 2002), both of which are crucial for prescriptive claims to be valid. If the causal relationships observed cannot be generalized to new

contexts, they cannot form a basis for recommendations for new contexts. This type of validity is thus related to the predictive power of a design principle's underlying causal model.

In the literature, external validity is often portrayed as concerning generalizations from a sample to a population, raising concerns about the difficulty in obtaining random samples in educational research (e.g., Marley & Levin, 2011). However, generalization from a sample to the population may not be the most relevant aspect of external validity for prescriptive claims for teaching. Teachers do not teach the population, but a group of students at a similar level as the groups of students studied. The key question is thus rather whether the causal explanation can be generalized to students in the teacher's own class. External validity in the context of teaching thus depends more on demonstrating the robustness of the causal explanation across students, teachers, and contexts, than on the size and sign of average effects on samples (Berliner, 2002).

In addition, it is of equal importance to know to which extent group-level effects hold for individual students. An average positive effect of an element of teaching does not warrant a general recommendation of that element, if the effect is due to a large increase in learning for little more than half of the students, and an equally large decrease for the rest. External validity of design principles thus requires detailed analysis of variations over groups and individuals. For example, a common concern regarding TDP 1 and 2 is that only high-achieving students will learn better when their CMR is supported, while low-achieving students will learn less. Studies within LICR have therefore investigated learning gains for different groups of students and do so far indicate that low-achieving students have equally much to gain from engaging in CMR (e.g., Jonsson, 2014). To increase this kind of external validity, one may thus adopt a different logic than statistical tests of causal descriptions where subdivisions of the sample are advised against since they reduce statistical power (Shadish et al., 2002).

Relative benefit

Relative benefit concerns the extent to which the positive effects of changing to *E* outweigh any negative effects. Changing practice often entails challenges for teachers related to motivation, resources, competence and beliefs (Gregoire, 2003), and it is necessary to consider whether the change is positive, if it is prioritized, and whether the available resources are appropriate. In relation to TDP 2, one must consider that many teachers are used to helping students by describing solution methods. In addition, changing to supporting student's CMR is not just changing from a familiar to an unfamiliar practice but to an inherently more difficult practice, at least until the students have reached a higher level of competence and require less help. During a period, the change may also cause discomfort for the students. For a design principle to be valid, using *E* should not create larger problems than it solves, compared to alternatives. Wecker (2013) sums this up:

> Compared to all other known options that cause outcome [*Y*] with sufficient probability, the expected positive (side) effects, the expected negative side effects and the expected necessary effort for the implementation of educational intervention A as a whole are more favorable. (p. 7)

In a causal claim of the type "*X* causes *Y*" it is not a necessary validity requirement that *Y* is a better outcome or that *X* is a better way of reaching *Y* than what already exists. A distinctive feature of prescriptive claims is that they must point toward practices that are improved compared to previously known alternatives (Cobb et al., 2015). This means that there may already be ways to support students' achievement of *G* but that *E* does it by less negative consequences. This implies that the relative benefit criterion can be fulfilled by a better *E*, by reaching a new *G*, or by both. From a methodological perspective, determining the relative benefit of *E* is easier if the different effects, including attainment of *G*, are easily measurable. In practice, the situation is often complex and may require a reasoned assessment based on qualitative descriptions of the learning that occurred, and other consequences observed and reported by students and teachers.

Guiding power

Guiding power concerns the extent to which a design principle is consequential for instructional decisions. This restricts which valid causal explanations can form a basis for a design principle. More specifically, this concerns the *differentiation* of the particulars and the *range* of the causal explanation.

For design principles to be valid in the sense that they produce the desired results, their key constructs need to be described in a way that enables a teacher to differentiate enacting *E* from not enacting *E*, students engaging in *L* from not engaging in *L*, and attainment from nonattainment of *G*. That is, the constructs need to be described on a level that is consequential for instructional decisions. A valid design principle for teaching describes *E* in terms of properties that are within the control of teachers, not only functions or desired consequences of *E* for the students. Without concern for differentiation, construct validity could be high in TDP 2 as "feedback that supports students' ability and responsibility" describes the feedback in terms of its desired functions. But this description does not aid teachers in devising such feedback, as it does not describe properties that a teacher can design beforehand. Valid design principles for teaching also describe *L* in terms of properties that are observable by teachers. This may put restrictions on how construct validity can be increased, for example, by disallowing measures that require specific instruments or expertise. For TDP 2, differentiation regarding *L* could be increased by specifying observable indicators of students' re-engagement in CMR. Differentiation is particularly important as we do not expect the causal explanation to be deterministic; *E* will not be guaranteed to result in *L*. For example, we cannot expect a particular type of feedback to result in CMR for every student in every situation, as students themselves and the surroundings also play parts in what the student will do. It will thus be crucial for the teacher to determine if their support was successful or if more support is needed.

For a design principle to be valid in the sense that is meaningful it needs range, meaning that *G* is a substantial part of what students are to learn, that *E* constitutes a substantial aspect of teaching, and that the causal explanation holds for a variety of situations and contexts. That is, the design principle should be sufficiently general to be consequential for instructional decisions. There are trade-offs between range and external validity. Demands on external validity could be met by restricting the design principle to the context in which the research underlying it was conducted, decreasing claims of generalization, but this would make the design principle meaningless as a recommendation for other teachers with other students at a later time. To optimize the usefulness of design principles, we argue that one should always aim to increase the range. For example, our first developments of TDP 2 were restricted to one feedback formulation for each main difficulty. This made it lack range, since it only provided guidance for situations when one feedback formulation was sufficient to reengage students in CMR, which was only true in about 40% of the cases. In subsequent iterations, we developed formulations and guidelines for follow-up feedback, increasing range (Säfström et al., 2025). However, the range is conversely restricted by demands on external validity. In the case of TDP 1, the range is large: all students, all problems, and mathematical competence in general. If there is no sufficient support for this range, TDP 1 would lack external validity.

Employing EDR to increase validity of design principles

As described above, EDR has been characterized as collaborative, interventionist, and iterative, entailing phases concerned with hypothesizing, empirical experimentation, and analysis. In this section, we aim to strengthen EDR as a methodology by going further than describing that these characteristics are present to describe how they can be employed to increase the validity of design principles for teaching.

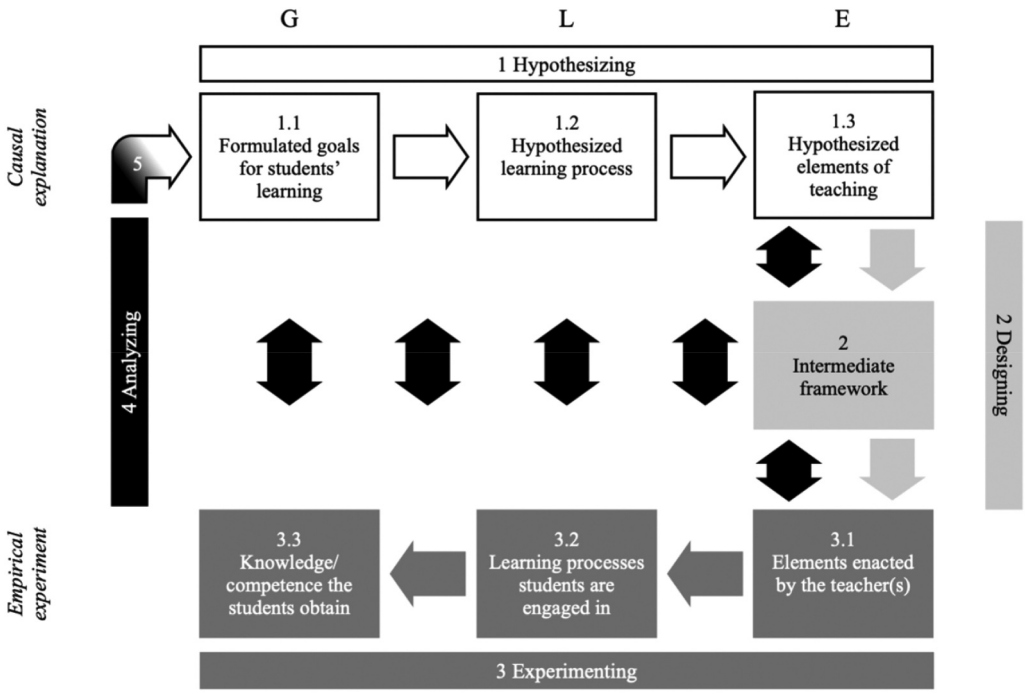


Figure 2. A process model for development of design principles within EDR.

We describe the EDR process in five main phases, summarized in Figure 2: *hypothesizing*, *designing*, *experimenting*, *analyzing*, and *iterating*. The rows of Figure 2 are based on Figure 1, with the upper row representing hypothetical relationships between constructs *G*, *L*, and *E* and the lower row representing real relationships between particular *G*, *L*, and *E* in an empirical experiment. In the following sections, we will describe how the five phases can be employed to increase the validity of design principles and explicate the precise role played by theory in each phase. We will use our own development of TDP 2 to exemplify and concretize how we work to increase the validity of research results in our own EDR projects.

Phase 1: hypothesizing

Before conducting empirical experiments, one should formulate an initial design principle (white boxes and arrows in Figure 2). This entails explicating the goal for student learning (1.1, Figure 2), hypothesizing a causal explanation for what learning processes promote the attainment of this goal (1.2, Figure 2), and hypothesizing what properties of elements of teaching promote those learning processes (1.3, Figure 2). There are no strong demands on the means used to formulate the initial design principle as it is hypothetical, and the ambition is to develop it. One can use, for example, pilot tests, local expertise, practical experience, and common sense (McKenney & Reeves, 2018). However, the research effort is more likely to succeed and avoid unnecessary detours if the initial design principle is grounded in the previous research and existing, relevant theories. Even if the initial design principle is vague, it is crucial to engage in hypothesizing before experimentation because it increases covariation validity: it is less likely for a specific prediction to be fulfilled by chance, than for a spurious correlation to exist in your data.[2]

To aim for relative benefit, the goal should be one that is not yet reached, generally, sufficiently or easily, by current teaching. If a design principle can be developed for such a goal, some negative effects, such as increased effort, might be more acceptable. To aim for range, the goal should constitute

a substantial part of important, desired knowledge, and the hypotheses about learning processes and elements of teaching should aim to cover a substantial subset of the situations that teaching and learning may entail. TDP 1 and 2 do so by covering properties of tasks and teacher support during task solving-two key elements of mathematics teaching. They are, nonetheless, focused as they exclude, for example, tests, whole-class discussions, and peer-to-peer interaction.

The hypotheses about salient features of design and expected learning processes and outcomes guide and focus on subsequent phases. In this sense, they structure the messy nonlaboratory settings and many dependent variables (Collins et al., 2004) entailed in EDR and function as a tool for "teasing complex interactions apart" (Kelly, 2004, p. 120).

Phase 2: designing

The initial hypotheses are usually not precise enough to directly guide instructional decisions in the moment. Therefore, the designing phase needs two steps (light gray boxes and arrows, Figure 2): constructing an intermediate framework and using the intermediate framework to guide enactment of teaching. Researchers' involvement in the first step is crucial, especially in early iterations when the design principle is vague. Teachers' involvement in at least the second step is crucial for testing and evaluating differentiation in later phases.[3]

The intermediate framework should be guided by the design principle, but in the early stages of a project the design principle may be very vague and general and the intermediate framework may be very concrete and specific, placing them rather far apart. To ensure that the design principle and the intermediate framework are connected, though it may initially be in implicit and vague ways, it is therefore important that those who formulated the initial design principle – which may often be the researchers initializing the project – are involved in the formulation of the intermediate framework. A key intermediate framework for the development of TDP 2 is a teacher guide for using formative assessment when students need help during problem solving. The first drafts of the teacher guide characterized students' specific reasoning difficulties in terms of lacking problem-solving competencies in relation to problem-solving phases, based on Schoenfeld's (1985) framework, and suggested feedback for each combination of competency and phase (Sidenvall et al., 2022). This can be seen as a specific way of using formative assessment, that is only one way of implementing the design principle. The intermediate framework may thus include more specific constructs than the design principle.

The intermediate framework can also be seen as a prototype of a practical result, as it is the intermediary between the design principle and teachers' enactment of teaching. It is a product that is meant to elaborate and specify the design principle in a form that is more easily used by teachers. For example, the final version of TDP 2 may describe the properties of feedback that supports students' evaluation of incorrect solutions – which we have found to be explicating (a) that students should evaluate, (b) what the student should evaluate, and (c) what to use as a means for evaluation but (d) not how to carry out the evaluation – while the intermediate framework also includes specific feedback formulations with these properties that teachers can use in interactions with their students, such as "You say that: __. How can you check if that is in line with the information given in the problem?" (Säfström et al., 2025).

Phase 3: experimenting

Experimenting – meaning that something is manipulated, and the effects of these manipulations are observed – is widely accepted as key for causal inference (Gravemeijer & Cobb, 2006; Shadish et al., 2002). To test a causal description, it can be enough to devise two alternative *E*s on group level and measure *G* afterward. To develop design principles, it is crucial to conduct experiments in collaboration with teachers allowing for variation and to collect data on *E*, *L*, *G*, and their relationships on student level (dark-gray boxes and arrows, Figure 2).

Conducting experiments in collaboration with teachers is important for external validity and relative benefit. Compared to conducting laboratory experiments where researchers act as teachers, which is an artificial setting, conducting experiments with teachers in their classes is grounds for stronger arguments for generalizing to other teachers' classes and is more likely to detect meaningful positive and negative side effects.

If several teachers participate, they will naturally vary in their enactment of *E*. Properties of elements of teaching usually also vary within classes. For example, students are regularly given different tasks and different support during interactions. Collecting detailed data on student level for all key components of the causal explanation allows one to use this variation to increase internal validity, construct validity and external validity. These variations are thus not to be seen as a threat to validity, but rather as a source of data allowing fine-grained analysis of how properties of *E* affect properties of *L*. As the causal explanation includes descriptions of the processes $E \rightarrow L$ and $L \rightarrow G$, one should strive to observe these processes, for example, how specific feedback is received and used by the student. This is, however, normally easier to do for $E \rightarrow L$ than $L \rightarrow G$, since a large part of the process $L \rightarrow G$ may occur in students' minds. When direct observation is not possible, detailed data on student level can allow for identification of traces of the specific *L* in the way *G* is attained, for example, the application of a method that was imitated or constructed during learning in tasks on a post test. Data on student level are also necessary to assess external validity by determining whether the causal explanation holds for all participating students or to which extent it varies, that is, if the general explanation can be generalized to individual students.

One might not test all components of the design principle in every experiment, or even every project. For example, TDP 2 is very complex, and studying all components and relationships in each experiment would be unfeasible. Since previous LICR projects had extensively studied the relationship $L \rightarrow G$, some of our projects focused on the relationship $E \rightarrow L$, allowing for greater variation in teachers, grades, and problems (Säfström et al., 2024, 2025). In an ongoing project, we include all components but focus on a smaller number of mathematical topics in grades 4–6.

Phase 4: analyzing

Rigorous analysis is necessary in all kinds of research, but to develop design principles three aspects are key: (a) the intermediate framework should be used as an analytic framework by (b) multiple coders in a (c) level-raising analysis (Simon, 2019) where the result at each level is used as the data at the next level, making inferences close to the data at each level. Using the intermediate framework as an analytic framework is crucial because the purpose of the analysis is not to understand what happened in the experiment per se but to assess the fit between what happened and the current version of the causal explanation to identify needs for development. This is done by conducting analysis on at least four levels, starting at coding of particulars, through categorization and then thematization, and finally assessing the fit between data and the current version of the intermediate framework (black boxes and arrows, Figure 2).

The first level is coding of each particular of the experiment (primarily instances of *E*, *L*, and *G*, indicated by the black double arrows between *E*, *L*, and *G* boxes in Figure 2), using the current version of the constructs included in the intermediate framework. At this level, multiple coders are key for generating data on the match between the constructs of the intermediate framework and the particulars of the experiment. The coding does not strive for agreement between coders but to reveal alternative interpretations of particulars in terms of the constructs in the intermediate framework. It is therefore crucial to use the intermediate framework as an analytic framework for the coding and to include multiple coders that may make independent interpretations. To aid analysis at the next level, it is important that coders argue for their coding as part of the analysis. To allow for assessment of differentiation, which concerns the design principle's ability to guide teachers, it is key that teachers partake in the coding. As some experiments can be restricted to focus certain components of the causal explanation, so will some analyses. For example, we first focused on diagnosis of student difficulties

during five iterations (Säfström et al., 2024), then the relationship between diagnosis, feedback, and student response during three iterations (Säfström et al., 2025). In the second study, we coded each feedback formulation and each student response. Detailed descriptions of how level-raising analysis was used and how it led to increased differentiation and range in relation to TDP 2 can be found in (Säfström et al., 2025).

The second level is categorization of the coding in terms of agreement and disagreement, resulting in a categorization of each particular as (a) a typical example of a construct, (b) an example matching no construct, or (c) a borderline case between several constructs. The coders' arguments are used to identify causes of agreements and disagreements, especially formulations in the intermediate framework that aid or obstruct reliable coding. From this level and up, the analysis is conducted by a few researchers that strive for agreement.

At the third level, the categorizations are analyzed across cases to discern patterns in the causes of agreement and disagreement, resulting in one or more themes for each construct. For example, during the development of the diagnostic framework (Säfström et al., 2024), this level resulted in one theme of indicators for each diagnosis and themes of issues regarding diagnoses that were not yet sufficiently well defined.

At the fourth level, the analysis shifts from the particulars to the relationships $E \rightarrow L$ and $L \rightarrow G$ and how well the current causal explanation fits the empirical data (indicated by the black double arrows between white and dark gray arrows in Figure 2). For $E \rightarrow L$, this means to assess the extent to which E is followed by L and the extent to which L is preceded by E, including quantitative measures but also qualitative scrutiny of the cases falling outside of $E \rightarrow L$ for possible additional properties required for E to be functional and alternative Es that also, and perhaps better, support L. The result of the level-raising analysis would then be an assessment of how well the intermediate framework and the causal explanation describe the E that supports L and the L that supports G and distinguishes such E and L from those that do not.

Phase 5: iterating phases 1–4

While each application of each phase can contribute to increasing the validity of design principles, the strength of EDR as a method for developing causal explanations and prescriptive claims lies in the iteration of the four phases (the black-to-white shaded arrow 5, Figure 2). The results of the analyzing phase should guide the revision of the causal explanation in the next round of the hypothesizing phase, increasing internal validity, construct validity, and differentiation by revising or further explicating constructs to better explain the gathered empirical data on the relationships $E \rightarrow L$ and $L \rightarrow G$. The revised explanation is then tested in the next design phase regarding whether it better guides enactment, in the next experimenting phase regarding whether it better predicts the consequences of E, and in the next analysis phase regarding whether it further aids the coding of E, L, and G. The repeated revision of the design principle and the intermediate framework based on results from the analyzing phase alleviate the risks of being open to a multitude of suggestions in the designing phase, pointed out by Kelly (2004), as suggestions that are not supported by the data will be removed from the design principle and the intermediate framework in subsequent hypothesizing and designing phases.

The development may not result in completely confirmed or refuted causal explanations but go from less to more substantiated and from vague to more elaborated claims. In early iterations, the causal explanation and its connection to the intermediate framework are often vague. Even so, we argue that the most effective way forward is to formulate initial design principles and intermediate frameworks and develop them over iterations, so that the development of theory is supported by empirical backing. Employing the design and analysis phases as described above will result in a more precise causal explanation in the next hypothesizing phase, providing better guidance for the next version of the intermediate framework in the designing phase. We developed the teacher guide and TDP 2 over eight iterations, making them more elaborated, valid, and aligned. The final version of the teacher guide (Lithner et al., 2025) includes definitions of all key constructs, specific questions for

gathering information on students' specific reasoning difficulties, support for diagnosing such difficulties that allows one to select functional feedback, feedback formulations that teachers find easy to use and adapt, and guidelines for interpreting and following up students' responses.

While experiments within EDR can include randomization and control groups, the main contrasting cases consist of variations of E within experiments and between iterations. For example, a teacher may use several types of feedback within one interaction where the student and the problem are the same, or a teacher may use a new type of feedback in the same class for a different problem in the next iteration, while another teacher uses this new type of feedback for the original problem in a different class. In extended teacher-researcher collaborations, the causal claims resulting from EDR may thus rely more on the logic of within-group research designs than between-group research designs. An advantage of contrasts between iterations is that at the time of experimentation, each *E* was believed to be able to support *L*, and not something researchers or teachers implemented for the case of control. This can aid ruling out effects of expectations of teachers and students and thus increase construct validity, which is important as studies of teaching cannot be blinded to remove expectation effects. If the causal explanation is found to repeatedly make successful predictions, or if the development of it leads to consistently increasing successfulness of predictions, it makes it implausible that observed variations are due to chance or another perfectly matched cause, increasing covariation and internal validity (Shadish et al., 2002).

Results from previous iterations should also guide the selection of contexts and cases in the experimenting phase, aiming to increase external validity with maintained range. In our recent projects, we conducted experiments over several years with over 50 teachers and 1000 students in grades 1–12 at seven schools, for various mathematical topics and task types, and we found no fundamental differences regarding what type of feedback was supporting CMR over this variation. However, our data were limited regarding grade 1 students and students in large difficulties and did not include mathematics at university level. We are currently working on the final version of TDP 2 and developments of TDP 1 for specific learning goals.

Discussion

In this article, we aimed to make a methodological contribution to educational design research (EDR) by explicating how it can be employed for the purpose of developing valid design principles for teaching. We have used examples from our projects to describe how four established validity types for causal claims can be adapted to design principles and introduced two additional validity types for design principles: relative benefit and guiding power. We have also described how valid design principles can be developed by employing characteristics and phases of EDR in specific ways.

One key characteristic of EDR is that it is collaborative (McKenney & Reeves, 2018). In this article, we have argued that for developing valid design principles for teaching, it is crucial to involve teachers in the experimenting and analyzing phases. When teachers use an intermediate framework to guide their enactment of teaching and for coding empirical data, variation in the interpretation of key constructs is introduced. This variation is important data for analysis at higher levels that can guide the development of design principles to increase construct validity and range. We should stress that our explication of teachers' crucial role in these phases for the purpose of increasing validity of design principles should not be read as an argument for not involving teachers in other parts of the EDR process. There are several purposes for involving teachers in EDR, for example, building long-term relationships that strengthen both research and development over time, as has been argued elsewhere (McKenney & Reeves, 2018).

Two other key characteristics of EDR are that it is interventionist and iterative (McKenney & Reeves, 2018). As described above, existing descriptions of these characteristics and phases in the process of EDR mostly focus on how experiments and iteration contribute to the development of the design (e.g., Bannan-Ritland, 2003; Easterday et al., 2018; Hoadley & Campos, 2022). If theory

development is described, it is mainly placed in a retrospective phase after iterations are completed (e.g., Cobb et al., 2017). In our process model, theory development is foregrounded within iterations and the role of theory in the five phases explicated. In particular, revising the causal explanation and letting it guide the design phase and predict outcomes in the next experimenting phase is crucial for testing its guiding and predictive powers, which is central for validity of the prescriptive claim. The model builds on the work of Cobb and colleagues (Cobb & Gravemeijer, 2008; Cobb et al., 2003, 2015, 2017; Gravemeijer & Cobb, 2006) and incorporates features they have advocated as key for theory development: aiming for generalizability, collecting data on students' learning processes, and starting with a goal for learning. The importance of starting with the goal is stressed in our models (Figures 1 and 2), which place the goal first, in contrast to other models, such as conjecture mapping (Sandoval, 2014).

We have also described additional aspects that are crucial for the development of valid design principles, explicating what it can mean to "put theory in harm's way" (Cobb et al., 2003). Figure 2 emphasizes the comparison between the causal explanation and the empirical test, stressing that the same components and relations must be regarded in both (indicated by the black double arrows). At the start, the upper row represents a possibly vague and less substantiated causal explanation from which an intermediate framework is more intuitively produced. The intermediate framework serves as both a guide for the enactment of teaching and as an analytic framework at the first level of analysis, focusing the particulars *E*, *L*, and *G*. Subsequent levels of analysis allow for the assessment of fit between the design principle, the intermediate framework, and what happened in the empirical experiment. Results of these assessments are used to revise the design principle, which in turn is used to revise the intermediate framework, aiming for them to increasingly better guide the designing phase and support enacted teaching that is more likely to support attainment of the desired goal. Iterating this process will develop both the design principle and the intermediate framework, making them more refined and strengthening their connection.

A key feature in our model is that we do not make a fundamental distinction between higher and lower levels of the theory. This sets our model apart from Sandoval's (2014) model, which places a higher level conjecture outside development and from EDR aiming to develop local instructional theories (DiSessa & Cobb, 2004; Prediger et al., 2015), which describe higher level theory as something that can inspire designs but is too general to guide it and to be tested. We refrain from making such a distinction for two reasons. First, we would argue that the level of theory developed by an EDR endeavor is a matter of range and the extent to which one convincingly link each experiment to a general causal explanation and devise experiments that reflect enough variation to build external validity that matches that range. If the causal explanation is a general theory, the strength in the connections between the general theory, the intermediate framework, and the enacted design determines whether experiments will test the general theory, only the intermediate framework or only the specific enactment. External validity of the resulting design principle is then an empirical question: experiments on division of fractions, geometrical constructions, and statistical distributions could all be "paradigmatic cases" of learning mathematical concepts if common properties of learning processes and elements of teaching are detected. Developing high-level theory could thus be done with the same EDR methods but would take more time and require a deliberate research program. Second, descriptions of high-level theory as "grand theory" that is inadequate to guide design and cannot be tested (DiSessa & Cobb, 2004; Prediger et al., 2015) raises questions about the value and role of such theories. If theory does not have a meaningful impact on how research is carried out, it is unclear why it is invoked (Niss, 2007).

The idea of aiming for increased range without losing external validity offers a different perspective on context, compared to recent calls for limiting claims of generality and restricting results to the contexts studied (e.g., Hirsh et al., 2022; Kaplan et al., 2020). Undoubtedly, being as transparent as possible about one's methodological choices is always important for critical review and interpretation of results, and it is good practice to explicitly argue for the proposed generality of one's results based on these choices. However, we argue that when the aim is developing design principles, it is important

to strive to increase their range by careful selection and consideration of context in each iteration and building explicit arguments for generalization from the contexts studied. In the end, one may realize that specification of context is necessary to reach a reasonable level of external validity and choose to publish a context-specific design principle. But including "in context *Z*" as a standard part of the generic design principle format, as suggested by van den Akker (2010), may suggest that design principles can be results of single teaching experiments as long as they include a precise description of the context where the experiment was conducted. This, we argue, would forfeit the meaning of design principles.

We agree with Dede (2004), Kelly (2004), and Shavelson et al. (2003) that the complexity and less well-defined constructs of EDR necessitates formulation of an explicit, transparent, and robust process specifying how and why claims are developed by systematic experiments and analyses and not based on unintended or biased grounds. We hope to have contributed to such a formulation in this article. A limitation of our contribution is that it is empirically grounded in work for one, albeit central, purpose of EDR: design principles for teaching. This type is at least partly distinct from other types of EDR, for example, one-on-one studies focusing on learning processes, curriculum development, professional development studies, and organizational studies (Cobb et al., 2015). It is an open question if the approach we describe can be used in other types of EDR. It seems reasonable, in line with how Cobb et al. (2015) delineate their characteristics of design research for various types, that the structure can be used generally but that the specific implementation may vary substantially between different types. For example, while the hypothesizing phase would still start by specification of the goal, the goals may be very different and formulated by different processes, and while relative benefit is still considered important for validity, it may entail other considerations, such as large-scale implementation costs and group-level effects. The relative importance of different aspects of validity may also vary, for example, making generalizability to a population more important for recommendations of large-scale implementation of educational policy.

Mathematics education research has been criticized for not sufficiently supporting improvement of practice (Stylianides & Stylianides, 2013). However, supporting such improvement is no simple research task. Guidance for practice does not follow directly from causal claims (Wecker, 2013), single experiments – no matter how well they are designed – are not enough to produce valid causal claims (Shadish et al., 2002), and meta-analyses of experiments often produce inconsistent results and inadequate guidance (Renkl, 2022). Empirically tested theories may provide better guidance for teachers (Renkl, 2022). However, if such theories are merely explanatory, they may still not provide sufficient guidance. Design principles are a specific form of prescriptive theory that may provide such guidance. We have considered what research efforts may be needed for developing valid design principles. Validity must be a basic criterion for design principles to be functional for teachers, but there may be additional criteria, for example, regarding their form. This is beyond our scope, but the process we describe requires the joint development of an intermediate framework that elaborates and specifies the design principle and acts as a guidance for and a tool for interpretation of teaching for teachers. It includes both concrete suggestions to test in one's practice and underlying explanations of how and why these suggestions will have specific outcomes. The final intermediate framework therefore has the potential of being directly useful as guidance for teachers in both changing and reflecting on their practice. However, the intermediate framework can be seen as a by-product of the process that is not the focus of development. Developing insights regarding if intermediate frameworks are a good way to disseminate design principles to teachers or what properties make them the most useful would require its own research efforts.

Our proposed format for design principles (Figure 1) gave rise to six types of validity. Fulfilling these six types is a high demand, especially since there are trade-offs between them. On the one hand, high demands are reasonable, for one should not make recommendations for practice lightly, as changes of practice always require effort and can have adverse effects. All research efforts may not warrant making prescriptive claims, even if they are rigorous and valuable in other ways. Just as ambitious learning goals require ambitious instructional practices (Cobb et al., 2018), we believe

ambitious goals for educational impact require ambitious methodologies. We argue that EDR is such a methodology. On the other hand, it is not reasonable to deem a design principle useless until it is completely and unambiguously elaborated and confirmed. The complexity of the relationships between teaching and learning, and their dependence on the will and actions of students and their peers, means that we will likely never reach deterministic causal descriptions and completely confirmed claims. Instructional decisions need to be made in the meantime (Harris, 2013). One should not see validity typologies as absolute criteria, but rather as heuristic devices for methodological decisions, reflection on the strengths and weaknesses of one's results, and for identification of needs for future research (Shadish et al., 2002). Used in this way, they can aid the building of arguments and empirical backing over time, resulting in increasingly accurate, explicated, elaborated, and sometimes refuted or radically changed causal explanations and recommendations for practice.

Notes

1. This aspect of construct validity is partly overlapping with internal validity and closely linked to what others have referred to as explanatory power.
2. See also, Brown's (1992) discussion on specific predictions for ruling out the Hawthorne effect.
3. Note that we use the terms "researcher" and "teacher" as labels for roles within a collaborative research project, where a researcher is someone conducting research work within the project, e.g., writing research plans, research project leadership, developing and using methods for analysis, and writing research reports, and a teacher is someone conducting teacher work within the project, such as planning, conducting, and reflecting on teaching. Project members may have multiple roles, and the roles are not necessarily determined by education or experiences. For example, most educational researchers in Sweden had a teacher degree and experience as practicing teachers when they entered research education, and there are practicing teachers with Ph.D. degrees.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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